

Bank Web Service Integration FSD

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1 Overview

1.1 An Overview

Airtel Money offers a mobile money platform that delivers mobile financial services to its customers. This platform enables subscribers to link their Airtel money wallets to their bank account and transact just like they would over the counter. This feature gives the customer the convenience of accessing their accounts over their phone and gives banks a wider reach of their services while reducing cost on infrastructure.

1.2 Scope

This document serves to describe the different services, their associated business rules and technical flows between the different systems being used within Airtel for bank integration.

Bank interfaces vary from bank to bank. Traditionally banks have used ISO 8583 but there is a growing trend for banks to use web services. As a consequence, from our experience of developing various bank interfaces over time and across different banks we have developed a standard web service API for bank integrations.

This interface provides a rich set of features for bank integrations and as a consequence has quite a few flows within the system.

2 Functional Specification

2.1 Objective

The solution provided has two main groups of functions:

1. **Mobile initiated:** these are transfers initiated by subscribers and / or agents and interact with the bank.
2. **Bank initiated:** These activities are initiated by bank staff through their banking module. While these activities can also be performed by the Web interface of Mobiquity the bank can opt to use its own in-house solution for tellers. This has the benefit that the bank staff will have access to only one interface and resolves issues around access to Mobiquity by bank staff.

2.2 System Overview

The main Mobile / Mobiquity initiated functionalities are described below.

2.2.1 Mobile initiated

- **Bank Balance Enquiry:** This function allows the customer to check the balance of the account linked to his mobile number.
- **Bank Mini Statement:** This function allows the customer to see the last few transactions with respect to the bank account linked to his mobile number.
- **Wallet to Own Bank account:** this allows subscribers and agents to move funds from their wallet into their bank account. The transaction will debit the customer's wallet and credit the bank wallet on the Mobiquity side. In addition, on the Bank side, it will debit the Airtel Money pool/trust account and credit the subscriber's bank account.
- **Wallet to Other Bank account:** this allows subscribers/agents to move funds from their wallets to another bank account that is not linked but within the same bank. The transaction will debit the customer's wallet and credit the bank wallet on the Mobiquity side. On the Bank's side, it will debit the Airtel Money pool/trust account and credit the recipient's bank account.
- **Bank to Own Wallet:** this function will allow subscribers to move funds from their bank accounts to their wallet. On the bank's side, the transaction will debit the customer's account and credit the Airtel Money pool/trust account. On the Mobiquity side, it will be debit the bank's wallet and credit the customer's wallet.
- **Bank to Other Wallet:** this function will allow subscribers to move funds from their bank accounts to another subscriber's wallet. On the bank's side, the transaction will debit the customer's account and credit the Airtel Money pool/trust account. On the Mobiquity side, it will be debit the bank's wallet and credit the recipient's wallet.
- **Bank To Bank:** this allows the subscriber's to move funds from their bank account to another bank account within the same bank. This transaction only happens on the bank's side. The customer's account will be debited and the recipient's account credited.
- **Merchant Auto Sweep:** This is a Mobiquity feature that will be automatically performed periodically (the current default setting is to run it every night but is configurable). During auto-sweep, all the e-value in the merchant's wallet will be debited and the same amount credited to the bank's wallet. On the bank's side, the merchant's account will be credited and the Airtel Money pool/trust account credited. This reduces the need to manually move the funds from the platform to the merchant's account.

2.2.2 Bank initiated

The main Bank initiated functionalities are described below:

- **Bank Account Registration:** This functionality allows the Bank to register/link a bank account number to the subscriber's mobile number.
- **Agent Deposit (Automatic O2C):** This function allows the agent to deposit cash in the branch and receive e-value in their wallet. This transaction debits the bank teller's account and credits the Airtel Money pool/trust account. On the mobiquity side, the agent's wallet is credited and the bank's wallet is debited.

3 Detailed Use Cases

3.1 Bank Balance Enquiry

Use Case ID	Bank_Integration_Case_01		
Use Case Name	Bank balance enquiry		
Description	Transaction flow for enquiry balance of bank account linked to Airtel Money		
Actors	Subscriber / Channel User		
Platform	Mobiquity		
Trigger Event	Initiated by User either via STK or USSD.		
Related Use Cases	NA		
Pre Conditions	• The subscriber should be a registered AM user • The subscriber's bank account should be linked to his AM wallet • There should be an existing integration between the respective banking system and Mobiquity EIG		
Post Conditions	• The existing transaction charges for balance enquiry will be debited from the customer's wallet.		
Frequency of Use	Medium		
Exceptional Condition	• If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events			
	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option.	System display menu of available banking services
	2	Customer chooses Bank Balance option	System prompts user to enter PIN
	3	Customer enters PIN	System notifies user that their balance enquiry is being processed. If validation is not passed check Alternate flow 1
	4	N/A	EIG makes an API call to the Bank system for balance enquiry on the customer's account
	5	N/A	Bank system receives the request, validates it and processes it and send a response to EIG with transaction status and the bank balance. If validation is not passed check Alternate flow 2

	6	N/A	Customer receives a notification of the bank balance and an SMS of the same is sent
	7	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 3: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.		
Alternate Flow 2	Alternate Flow 2 starts from Basic Flow step 5: Bank business validation failure was encountered during transaction processing. 1. The bank responds to Mobiquity with the failure status 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends		
Business Rules	1. All transactions will be done in the local currency. 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber, agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. SMS messages are configurable		

3.2 Wallet to Own Bank account

Use Case ID	Bank_Integration_Case_02
Use Case Name	Transfer from AM wallet to own bank account
Description	Transaction flow for funds transfer from a subscriber's wallet to their bank account
Actors	Subscriber / Channel User
Platform	Mobiquity
Trigger Event	Initiated by user either via STK or USSD.
Related Use Cases	NA
Pre Conditions	• The subscriber should be a registered AM user. • The subscriber's bank account should be linked to his AM wallet. • The subscriber's wallet should have enough funds for the required transfer. • There should be an existing integration between the respective banking system and Mobiquity EIG.
Post Conditions	• The existing transaction charges will be debited from the customer's wallet • The subscriber's wallet will be debited with the amount and his bank account credited

Frequency of Use	High		
Exceptional Condition	If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses From Airtel Money option	System displays list of available options
	3	Customer chooses 'To own bank a/c' option	System prompts user to enter amount in local currency
	4	Customer enters amount	System displays message of amount to be sent to own account and prompts user to confirm
	5	Customer confirms	System prompts for PIN
	6	Customer enters PIN and confirms	System notifies customer that their wallet to own bank account request is being processed.
	7	N/A	Mobiquity does the respective verification and validation on the request. If validation is passed, the customer's wallet is debited and the bank's wallet is credited. If validation is failed, check Alternate flow 1
	8	N/A	EIG makes an API call to the Bank system with the transaction details
	9	N/A	Bank system receives the request, validates, credits the customer account and debits the Airtel Money pool/trust account. A response is sent to EIG with transaction status If validation is not passed check Alternate flow 1
	10	N/A	System notifies the customer that the amount was successfully sent to their bank account. An SMS with the details will also be sent to the subscriber
	11	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 7: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are		

	made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Alternate Flow 2	Alternate flow 2 starts from Basic flow step 9: Bank system validation failure was encountered during transaction processing: 1. The system marks the transaction as failed and reverses the debit/credit on Mobiquity side 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Business Rules	1. All transactions will be done in the local currency. 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. SMS messages are configurable

3.3 Wallet to Other Bank account

Use Case ID	Bank_Integration_Case_03				
Use Case Name	Transfer from AM wallet to other bank account				
Description	Transaction flow for funds transfer from a subscriber's wallet to other bank account				
Actors	Subscriber / Channel User				
Platform	Mobiquity				
Trigger Event					
Related Use Cases					
Pre Conditions	• The subscriber should be a registered AM user • The subscriber's bank account should be linked to his AM wallet • The subscriber's wallet should have enough funds for the required transfer • There should be an existing integration between the respective banking system and Mobiquity EIG				
Post Conditions	• The existing transaction charges will be debited from the customer's wallet • The beneficiary's wallet will be debited with the amount and subscriber's bank account will be credited				
Frequency of Use	High				
Exceptional Condition	• If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.				
Basic Flow of Events	<table> <tr> <th>S.No.</th><th>User Action</th><th>System Response</th></tr> </table>		S.No.	User Action	System Response
S.No.	User Action	System Response			

	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses From Airtel Money option	System displays list of available options
	3	Customer chooses 'To other bank a/c' option	System prompts user to enter beneficiary's first name
	4	Customer enters beneficiary's first name	System prompts for beneficiaries second name
	5	Customer enters beneficiary's second name	System prompts for beneficiary's account number
	6	Customer enters account number	System prompts for amount
	7	Customer enters amount	System displays message of amount to be sent to beneficiary and their account number and prompts user to confirm
	8	Customer confirms	System prompts for PIN
	9	Customer enters PIN and confirms	System notifies customer that their wallet to other bank account request is being processed.
	10	N/A	Mobiquity does the respective verification and validation on the request. If validation is passed, the customer's wallet is debited and the bank's wallet is credited. If validation is failed, check Alternate flow 1
	11	N/A	EIG makes an API call to the Bank system with the transaction details
	12	N/A	Bank system receives the request, validates, credits the beneficiary account and debits the Airtel Money pool/trust account. A response is sent to EIG with transaction status If validation is not passed check Alternate flow 2
	13	N/A	System notifies the customer that the amount was successfully sent to the beneficiary bank account. An SMS with the details will also be sent to the subscriber and the beneficiary
	14	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step10: Business Validation failure was encountered during transaction processing.		

	1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Alternate Flow 2	Alternate flow 2 starts from Basic flow step 12: Bank system validation failure was encountered during transaction processing: 1. The system marks the transaction as failed and reverses the debit/credit on Mobiquity side 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Business Rules	1. All transactions will be done in the local currency 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. The recipient bank account should be in the same bank as the subscriber's. 7. In the case of sending money to another bank account the recipient will also receive a message. 8. SMS messages are configurable

3.4 Bank to Own Wallet

Use Case ID	Bank_Integration_Case_04
Use Case Name	Transfer from subscriber's bank account to their wallet
Description	Transaction flow for funds transfer from bank account to their wallet
Actors	Subscriber / Channel User
Platform	Mobiquity
Trigger Event	
Related Use Cases	
Pre Conditions	• The subscriber should be a registered AM user • The subscriber's bank account should be linked to his AM wallet • The subscriber's bank account should have enough funds for the required transfer • There should be an existing integration between the respective banking system and Mobiquity EIG
Post Conditions	• The existing transaction charges will be debited from the customer's wallet • The subscriber's wallet will be credited with the amount and his bank account debited
Frequency of Use	High

Exceptional Condition	If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses From Bank Account option	System displays list of available options
	3	Customer chooses 'To own Airtel Money wallet' option	System prompts user to enter amount in local currency
	4	Customer enters amount	System displays message of amount to be sent to their wallet and prompts user to confirm
	5	Customer confirms	System prompts for PIN
	6	Customer enters PIN and confirms	System notifies customer that their bank to own wallet request is being processed.
	7	N/A	Mobiquity does the respective verification and validation on the request. If validation is passed, the request is passed on to EIG. If validation is failed, check Alternate flow 1
	8	N/A	EIG makes an API call to the Bank system with the transaction details
	9	N/A	Bank system receives the request and validates it. If validation is passed, the customer's account is debited and the Airtel Money pool/trust account is credited with the amount. A response is sent to EIG with transaction status If validation is not passed check Alternate flow 2
	10	N/A	If the response received from the banking system is successful, Mobiquity will debit the bank's wallet and credit the customer's wallet with the amount
	11	N/A	System notifies the customer that the amount was successfully sent to their wallet. An SMS with the details will also be sent to the subscriber
	12	The use case ends.	

Alternate Flow 1	Alternate Flow 9 starts from Basic Flow step 7: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Alternate Flow 2	Alternate flow 2 starts from Basic flow step 9: Bank system validation failure was encountered during transaction processing: 1. The system marks the transaction as failed and no charges are made to the subscriber's wallet 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.
Business Rules	1. All transactions will be done in the local currency 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grades/Categories of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. The recipient bank account should be in the same bank as the subscriber's 7. In the case of sending money to another bank account the recipient will also receive a message. 8. SMS messages are configurable

3.5 Bank to Other wallet

Use Case ID	Bank_Integration_Case_05
Use Case Name	Transfer from subscriber's bank account to other wallet
Description	Transaction flow for funds transfer from bank account to other wallet
Actors	Subscriber / Channel User
Platform	Mobiquity
Trigger Event	
Related Use Cases	
Pre Conditions	• The subscriber should be a registered AM user • The subscriber's bank account should be linked to his AM wallet • The subscriber's bank account should have enough funds for the required transfer • There should be an existing integration between the respective banking system and Mobiquity EIG
Post Conditions	• The existing transaction charges will be debited from the customer's wallet • The beneficiary's wallet will be credited with the amount and the subscriber's bank account debited
Frequency of Use	High

Exceptional Condition	If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses From Bank Account option	System displays list of available options
	3	Customer chooses 'To other Airtel Money wallet' option	System prompts user to enter beneficiary's number
	4	Customer enters beneficiaries number	System prompts user to enter amount
	5	Customer enters amount	System displays message of amount to be sent to beneficiary's number and prompts user to confirm
	6	Customer confirms	System prompts for PIN
	7	Customer enters PIN and confirms	System notifies customer that their bank to other wallet request is being processed.
	8	N/A	Mobiquity does the respective verification and validation on the request. If validation is passed, the request is passed on to EIG. If validation is failed, check Alternate flow 1
	9	N/A	EIG makes an API call to the Bank system with the transaction details
	10	N/A	Bank system receives the request and validates it. If validation is passed, the customer's account is debited and the Airtel Money pool/trust account is credited with the amount. A response is sent to EIG with transaction status If validation is not passed check Alternate flow 2
	11	N/A	If the response received from the banking system is successful, Mobiquity will debit the bank's wallet and credit the beneficiary's's wallet with the amount
	12	N/A	System notifies the customer that the amount was successfully sent to beneficiary's wallet. An SMS with the details will also be sent to the subscriber and the beneficiary

	13	The use case ends.
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 8: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.	
Alternate Flow 2	Alternate flow 2 starts from Basic flow step 10: Bank system validation failure was encountered during transaction processing: 1. The system marks the transaction as failed and no charges are made to the subscriber's wallet 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.	
Business Rules	1. All transactions will be done in the local currency 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. The recipient bank account should be in the same bank as the subscriber's 7. In the case of sending money to another bank account the recipient will also receive a message. 8. SMS messages are configurable	

3.6 Bank to Bank

Use Case ID	Bank_Integration_Case_06
Use Case Name	Transfer from subscriber's bank account to other account
Description	Transaction flow for funds transfer from one bank account to other bank account
Actors	Subscriber / Channel User
Platform	Mobiquity
Trigger Event	
Related Use Cases	
Pre Conditions	• The subscriber should be a registered AM user • The subscriber should have enough balance in his wallet to cover the service charges • The subscriber's bank account should be linked to his AM wallet • The subscriber's bank account should have enough funds for the required transfer • There should be an existing integration between the respective banking system and Mobiquity EIG

Post Conditions	- The existing transaction charges for bank to bank transfer will be debited from the customer's wallet - The beneficiary's bank account will be credited with the amount and the subscriber's bank account debited		
Frequency of Use	High		
Exceptional Condition	If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses From Bank Account option	System displays list of available options
	3	Customer chooses 'To other Bank Account' option	System prompts user to enter beneficiary's first name
	4	Customer enters beneficiary's first name	System prompts user to enter beneficiary's second name
	5	Customer enters beneficiary's second name	System prompts user to enter beneficiary's account number
	6	Customer enters beneficiaries account number	System prompts user to enter amount
	7	Customer enters amount	System displays message of amount to be sent to beneficiary's account number from own account and prompts user to confirm
	8	Customer confirms	System prompts for PIN
	9	Customer enters PIN and confirms	System notifies customer that their bank to bank request is being processed.
	10	N/A	Mobiquity does the respective verification and validation on the request. If validation is passed, the request is passed on to EIG. If validation is failed, check Alternate flow 1
	11	N/A	EIG makes an API call to the Bank system with the transaction details
	12	N/A	Bank system receives the request and validates it. If validation is passed, the customer's account is debited and the beneficiary's account is credited with the amount. A response is sent to EIG with transaction status

			If validation is not passed check Alternate flow 2
	13	N/A	System notifies the customer that the amount was successfully sent to beneficiary's account. An SMS with the details will also be sent to the subscriber and the beneficiary
	14	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 10: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.		
Alternate Flow 2	Alternate flow 2 starts from Basic flow step 12: Bank system validation failure was encountered during transaction processing: 1. The system marks the transaction as failed and no charges are made to the subscriber's wallet 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.		
Business Rules	1. All transactions will be done in the local currency 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. The recipient bank account should be in the same bank as the subscriber's 7. In the case of sending money to another bank account the recipient will also receive a message. 8. SMS messages are configurable		

3.7 Bank Mini Statement

Use Case ID	Bank_Integration_Case_07
Use Case Name	Bank mini statement
Description	Request for mini statement of bank account linked to Airtel Money
Actors	Subscriber / Channel User
Platform	Mobiquity
Trigger Event	
Related Use Cases	

Pre Conditions	• The subscriber should be a registered AM user • The subscriber's wallet should have enough balance to cover the service charges • The subscriber's bank account should be linked to his AM wallet • There should be an existing integration between the respective banking system and Mobiquity EIG		
Post Conditions	• The existing transaction charges will be debited from the customer's wallet		
Frequency of Use	High		
Exceptional Condition	• If there is a timeout during the transaction, the ambiguous transaction workflow will be initiated and the user is informed that the transaction is in progress.		
Basic Flow of Events	S.No.	User Action	System Response
	1	Customer initiates Airtel money USSD/STK menu and chooses bank option	System display menu of available banking services
	2	Customer chooses Bank Statement option	System prompts user to enter PIN
	3	Customer enters PIN	System notifies user that their bank statement request is being processed. If validation is not passed check Alternate flow 1
	4	N/A	EIG makes an API call to the Bank system for mini statement request on the customer's account
	5	N/A	Bank system receives the request, validates, processes it and sends a response to EIG with transaction status and the bank balance. If validation is not passed check Alternate flow 2
	6	N/A	Customer receives a notification and details of the bank statement and an SMS of the same is sent
	7	The use case ends.	
	Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 3: Business Validation failure was encountered during transaction processing. 1. The system marks the transaction as a failed transaction & no threshold changes are made to the subscriber's wallet. 2. Airtel money sends notification to the sender with the details of the failure condition. 3. Use case ends.	
Alternate Flow 2	Alternate Flow 2 starts from Basic Flow step 5: Bank business validation failure was encountered during transaction processing. 1. The bank responds to Mobiquity with the failure status 2. Airtel money sends notification to the sender with the details of the failure condition.		

	3. Use case ends
Business Rules	1. All transactions will be done in the local currency 2. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 3. AntiMoney Laundering (Transfer Control Profiles) will apply depending on the Grade/Category of the subscriber or agent or super-agent. 4. Service charges and Commissions will also apply in a similar way. 5. The customer / agent / super-agent must have the entire value of the transaction, including the service charge in their wallet or bank account to complete the transaction. 6. SMS messages are configurable

3.8 Bank Account Registration

Use Case ID	Bank_Integration_Case_08		
Use Case Name	Bank account registration		
Description	Linking of subscriber's/ channel user's wallet to a bank account		
Actors	Subscriber / Channel user		
Platform	Mobiquity / Bank platform		
Trigger Event			
Related Use Cases			
Pre Conditions	• The user should be a registered AM user • The user should have all the required KYC requirements • The bank should have an Airtel Money trust/pool account • There should be an existing integration between the respective banking system and Mobiquity EIG		
Post Conditions	• The user's wallet will be linked to a bank account		
Frequency of Use	High		
Exceptional Condition			
Basic Flow of Events			
	S.No.	User Action	System Response
	1	Customer goes to the bank to request for bank registration and provides all the KYC requirements.	Bank attendant / teller verifies the customer details and initiates the process of linking the customer's account with their mobile number.
	2	Bank verifies customer details and initiates linking with Mobiquity.	Bank system sends a request to the EIG which performs the subscriber

			validation. If the validation fails Alternate Flow 1 is followed.
	3	N/A	Mobiquity responds successfully and the bank informs the customer. SMS sent to customer.
	4	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 2: Business Validation failure was encountered during transaction processing. 1. The system responds to the bank system that registration cannot be completed. 2. Use case ends.		
Business Rules	1. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 2. KYC details are required to be presented by the customer.		

3.9 Automatic O2C

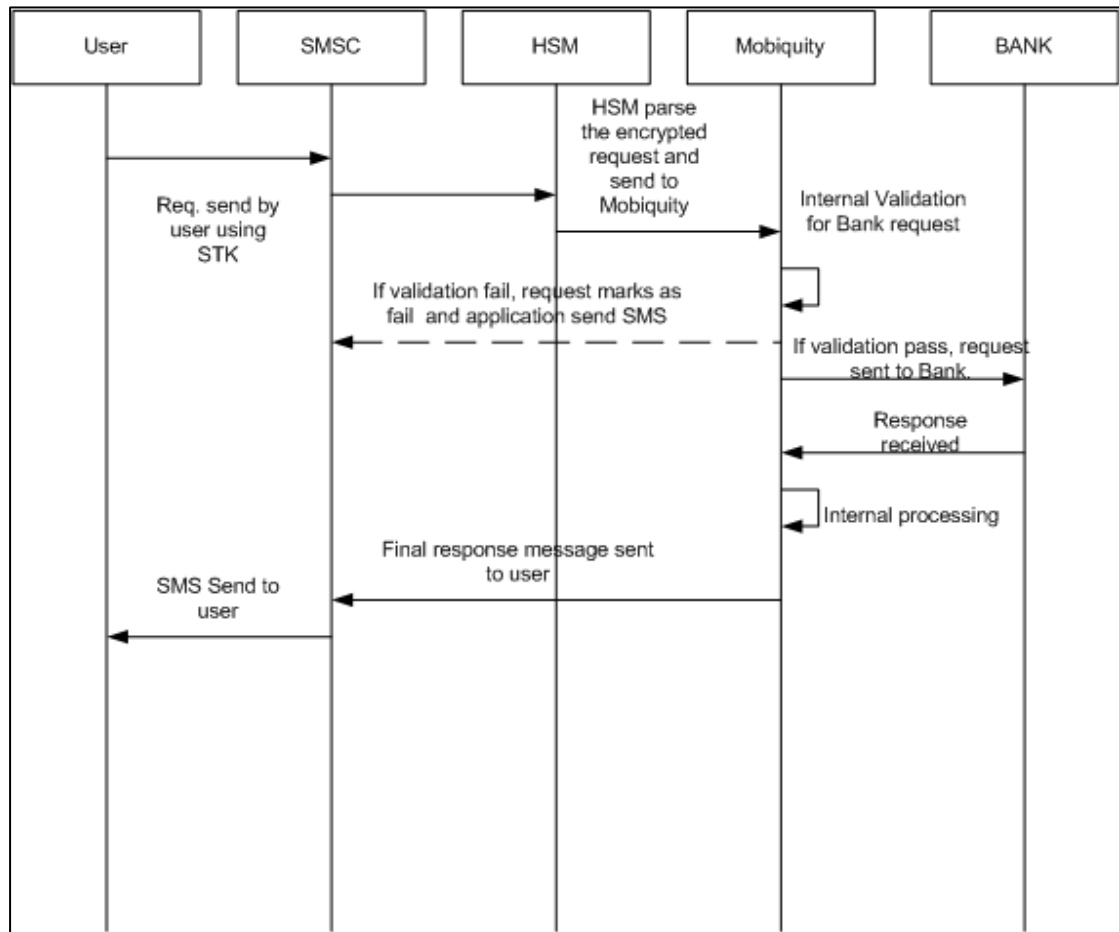
Use Case ID	Bank_Integration_Case_09		
Use Case Name	Automatic O2C		
Description	Depositing cash into agent/super-agent wallet following deposit in bank.		
Actors	Channel user		
Platform	Mobiquity / Bank platform		
Trigger Event			
Related Use Cases			
Pre Conditions	- The user should be a registered AM user - The user should have all the required KYC requirements - The bank should have an Airtel Money trust/pool account - There should be an existing integration between the respective banking system and Mobiquity EIG		
Post Conditions	The user's wallet will be linked to a bank account		
Frequency of Use	High		
Exceptional Condition			
Basic Flow of Events	S.No.	User Action	System Response
	1	Channel User goes to the bank to deposit funds into channel user wallet.	Bank attendant / teller verifies the customer details and initiates the process of deposit the customer's account with their mobile number.

	2	Bank verifies customer details and initiates O2C with Mobiquity.	Bank system sends a request to the EIG which performs the subscriber validation. If the validation fails Alternate Flow 1 is followed.
	3	N/A	Bank system debits the bank teller's bank account and credits the Airtel Money pool/trust account. Mobiquity credits the channel user wallet and debits the bank wallet.
	4	N/A	Mobiquity responds successfully and the bank informs the customer. SMS sent to customer.
	5	The use case ends.	
Alternate Flow 1	Alternate Flow 1 starts from Basic Flow step 2: Business Validation failure was encountered during transaction processing. 1. The system responds to the bank system that registration cannot be completed. 2. Use case ends.		
Business Rules	1. The service will be available to all subscribers, agents and super-agents of all Grades/Categories. 2. KYC details are required to be presented by the customer.		

3.10 Sequence Diagrams

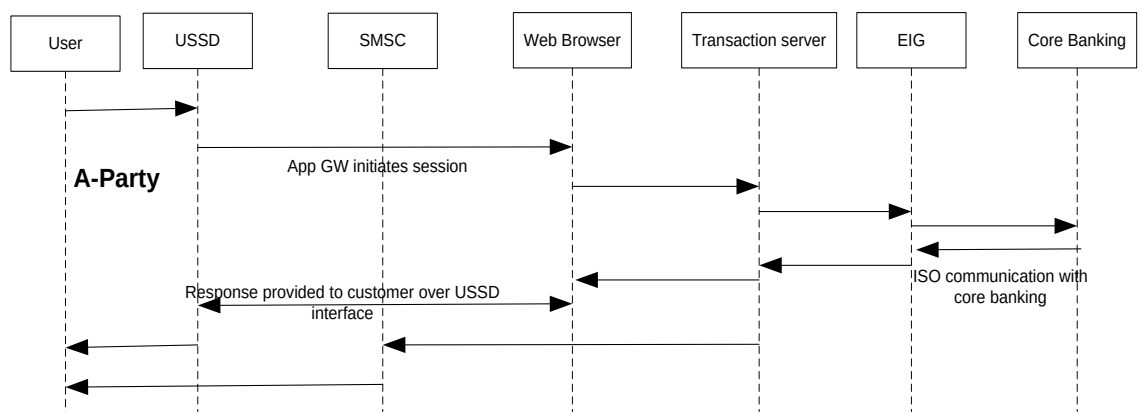
The sequence diagrams for Mobile initiated transactions are shown below:

- Sequence diagram for Mobile initiated transactions on STK menu:

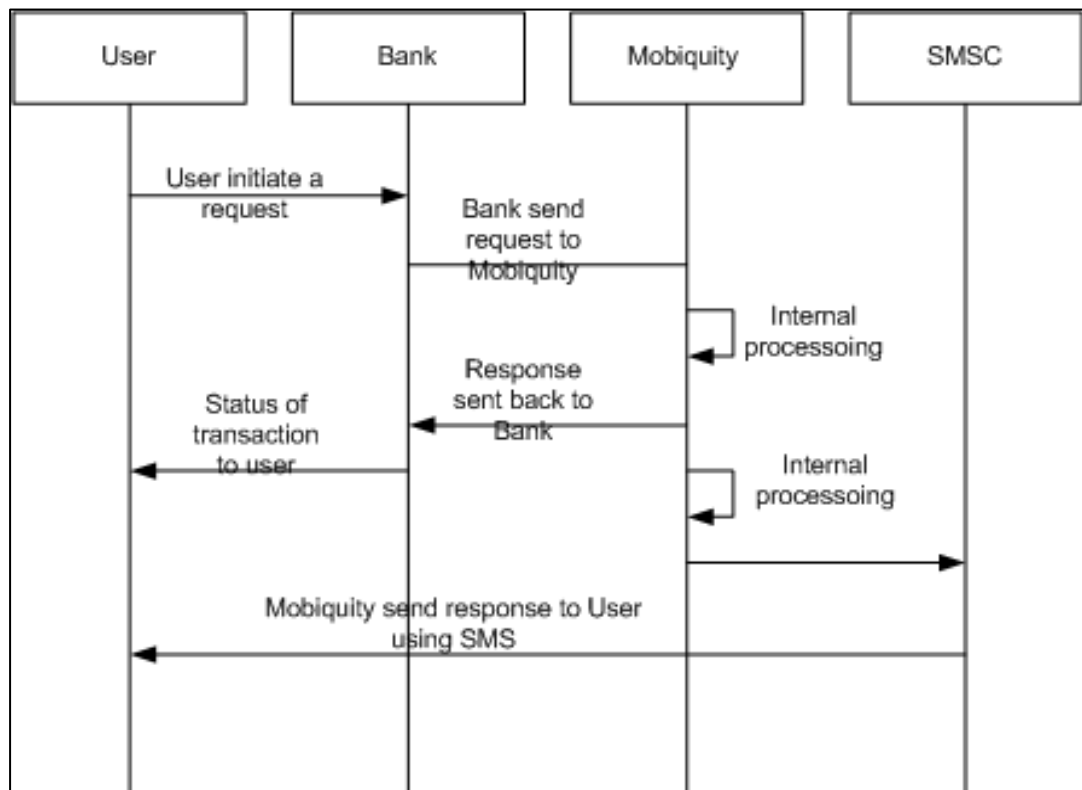


- Sequence diagram for Mobile initiated transactions on USSD menu:

Mobile Initiated Transactions

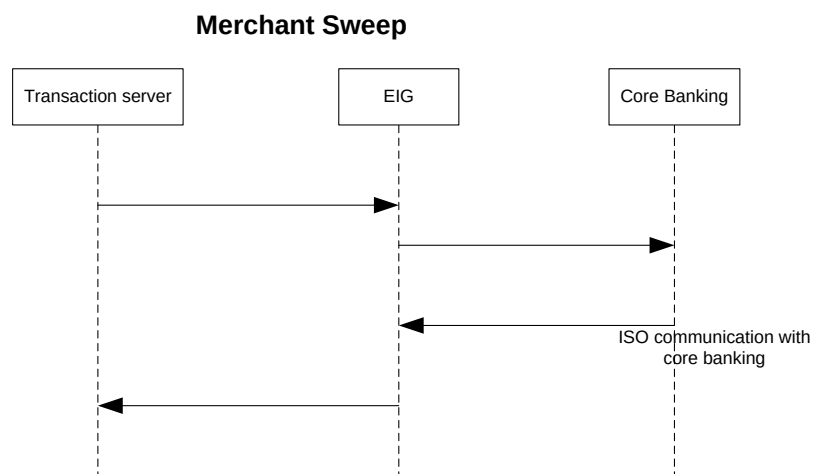


The sequence diagram for Bank initiated diagrams is as shown below:



3.11 Merchant Sweep

Merchant sweep has the following flow:



4 Technical Specifications(Web Service)

The integration between Mobiquity and the banking system will be through web service

4.1 Objectives

- Funds outflow: Movement of e-value from Mobiquity to own and other bank account. This also cover merchant sweep.
- Funds inflow: Receive e-value from bank account to Mobiquity and to other wallet.
- Bank Balance: To check the balance in his bank account.
- Bank Mini Statement: To check the mini statement of bank transactions.
- Bank initiated transactions: Account linking and automated O2C.

4.2 Connection Protocol

The Mobiquity™ is communicating with the Bank over HTTPS protocol. In this Mobiquity™ is sending request to Bank.

4.3 Bank WSDL Details

Sr. No.	Method Name	Input field	Output Field	Use Case #
1.	balanceEnquiry	txtBankName txtBranchCode txtAccountNumber txtBFName txtBLName txtRefID mydate txtCountryCode txtAuthNum txtExternal1 txtExternal2	balance status transactionAmount transactionDetails transactionID transactionTimeStamp	• Bank_Integration_Case_01
2.	fundTransferAccountToMobile	txtBankName txtBranchCode txtAccountNumber txtBFName txtBLName txtAmount txtCurrency txtRefID mydate txtCountryCode txtAuthNum txtExternal1 txtExternal2	balance status transactionAmount transactionDetails transactionID transactionTimeStamp	• Bank_Integration_Case_04 • Bank_Integration_Case_05
3.	fundTransferMobileTo	txtBankName	balance	• Bank_Integration_C

Sr. No.	Method Name	Input field	Output Field	Use Case #
	Account	txtBranchCode txtAccountNumber txtBFName txtBLName txtAmount txtCurrency txtRefID mydate txtCountryCode txtAuthNum txtExternal1 txtExternal2	status transactionAmount transactionDetails transactionID transactionTimeStamp	ase_02 • Bank_Integration_Case_03
4.	transactionEnquiry	txtBankName txtBranchCode txtAccountNum txtBFName txtBLName txtAmount txtCurrency txtRefID mydate txtCountryCode txtAuthNum txtExternal1 txtExternal2	balance status transactionAmount transactionDetails transactionID transactionTimeStamp	Bank_Integration_Case_07
5.	FundTransferAccountToAccount	dBankName dBranchCode dAccountNumber dFName dLName cBankName cBranchCode cAccountNumber cFName cLName txtAmount txtCurrency txtRefID mydate txtCountryCode smsSourceMsisdn txtExternal1 txtExternal2	balance status transactionAmount transactionDetails transactionID transactionTimeStamp	Bank_Integration_Case_06

4.4 Mobiquity™ WSDL Details

The following WSDL will be used to receive requests from the Bank.

Sr. No.	Method Name	Input field	Output Field	
1.	BankAccountRegistration	referenceid msisdn bankname branchcode accountnumber	ResponseCode ResponseDescription ReferenceID TimeStamp MSISDN	Bank_Integration_Case_08
2.	DepositToMobileMoney	referenceid msisdn transactionamount currencycode transactiondatetime	ResponseCode ResponseDescription ReferenceID TimeStamp MSISDN TransactionAmount	Bank_Integration_Case_09

5 Definitions, Acronyms & Abbreviations

AM	Airtel Money
MSISDN	Mobile Subscriber Integrated Services Digital Network Number (i.e. Mobile Number)
SMS	Short Message Service
USSD	Unstructured Supplementary Services Data
PIN	Personal Identification Number
EIG	External Interface Gateway
STK	Simcard Tool Kit
OAM	Operation, Administration and Maintenance
Mobiquity™	Brand Name of Comviva's mCommerce solution platform.
BP (Bill Payment)	Subscribers can pay utility bills directly from their mWallet.
MP (Merchant Payment)	Subscribers can pay for goods and services to merchants from their mWallet.

6 Document Change History

Version Number	Description of Changes	Author	Reviewer	Comments
v1.0	Initial Draft	David Kimani	Oscar Correia	
v1.1	Revision of document based on input from Bank document from Avinash.	David Kimani	Oscar Correia	
v1.2	Revision of document based on review from internal teams.	David Kimani	Oscar Correia	
v1.3	Review of functional specifications.	David Kimani	Oscar Correia	
v1.4	Review of document and correction of typos.	David Kimani	Oscar Correia	
V1.5	Review according to input of technical team	David Kimani	Oscar Correia	

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